

Exploring the development directions of Physics and General Relativity

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Abstract:

This paper examines the current challenges and future directions in general relativity. On one hand, it discusses issues such as black hole singularities within continuous models, as well as the mathematical inconsistencies in the construction of three-dimensional continuous space models, highlighting that investigating the adjacency relations of three-dimensional space should be a key focus for future physics research. On the other hand, based on the elastic spacetime paradigm in general relativity, this paper proposes that studying the global elastic coefficient function of space is essential for understanding the nature of spacetime.

Keywords: General Relativity; Space adjacency relations; global elastic coefficient function of space

1. General relativity[1] is established upon a continuous 3+1 dimensional spacetime framework.
2. The core of general relativity lies in the elastic spacetime paradigm, which involves metric variations and geodesic calculations.
3. General relativity posits that matter and its interactions influence the metric, which in turn governs the trajectories of matter.
4. The theory does not explicitly clarify whether matter and spacetime are unified. If unified, it naturally explains why mass-generated gravitational fields exhibit spherical symmetry. If treated as separate entities, however, there lacks an intuitive physical interpretation for this spherical divergence.
5. The mathematical model of continuous spacetime presents paradoxes such as black hole singularities.
6. Mathematically, continuous space is constructed by extending the real number set to three dimensions. While the real number set has a well-defined linear adjacency structure (e.g., Dedekind cuts or nested intervals), the adjacency relations for spherical neighborhoods in 3D space remain mathematically ambiguous, leading to conceptual imprecision.
7. Without a rigorously defined adjacency structure, the tensor networks and geodesic computations in general relativity remain approximate phenomenological fittings—whether in continuous or discrete models. Only by establishing explicit spatial adjacency relations can more precise and physically compelling equations be formulated. Considering the isotropy of spatial interactions, the overall adjacency distribution in space should exhibit a polycrystalline structure with periodic dislocations. How can such dislocations be rigorously defined?
8. The study of discrete adjacent configurations in space elementary quanta or

continuous fractal adjacency configurations with infinite downward subdivision represents in continuous model not only a pivotal direction for the next phase of general relativity development, but also constitutes a primary research trajectory for both theoretical and experimental physics. This is because adjacency configurations fundamentally determine physical properties and governing laws.

9. Beyond spatial adjacency, the elastic spacetime paradigm of general relativity suggests another key research avenue: given quark confinement and asymptotic freedom in QCD, as well as the nonlinear, asymmetric distribution of dark energy density in cosmological observations, the spacetime elasticity coefficient function may be piecewise, nonlinear, and asymmetric. Thus, studying this global elasticity function—spanning cosmology to high-energy particle physics—constitutes a foundational challenge and central direction for future physics research.[2][3][4][5]

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